Project Title: PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning **Project Design Phase-I** **TeamID:** 699d5f8fc40b5ad0e9ee59d5

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

- Farmers
- Home gardeners
- Agricultural students
- Nursery owners
- Agriculture researchers

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

- Poor internet connectivity
- Lack of technical knowledge
- Low digital literacy
- Limited smartphone capability

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

- Manual disease identification, consulting agricultural experts, Google image search, and farming books are commonly used methods for detecting plant diseases.
- These methods are **time-consuming and require expert knowledge** to correctly identify the disease.
- They **may not always provide quick or accurate diagnosis**, especially for early-stage plant diseases.

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

- Identify plant diseases early
- Prevent crop damage
- Get correct treatment recommendations
- Save time compared to manual inspection
- Reduce crop losses

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

- Many farmers and gardeners **lack proper knowledge to accurately identify plant diseases**.
- **Agricultural experts are not always easily accessible**, especially in rural areas.
- Traditional methods of diagnosis are **slow and may lead to incorrect treatment**, which can cause crop loss.

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Customers usually:

- Search plant disease images online
- Ask agriculture experts
- Use WhatsApp farming groups
- Look for YouTube farming solutions

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e.seeing their neighbor installing solar panels, reading about a more efficient solution in the news.

Customers act when:

- Plant leaves show unusual spots
- Crop productivity decreases

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control – use it in your communication strategy & design.

Before:

- Confused
- Worried about crop loss
- Lack of expert advice

After:

- Confident about diagnosis
- Better crop management

10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

PlantCare-AI

An AI-powered plant disease detection system that:

- Uses deep learning to analyze plant leaf images
- Identifies plant diseases automatically
- Provides treatment recommendations
- Helps farmers and gardeners maintain healthy plants

Key features:

- Image-based disease detection
- AI prediction model
- Disease description
- Treatment suggestions

8. CHANNELS of BEHAVIOUR CH

ONLINE

What kind of actions do customers take online? Extract online channels from #7

OFFLINE

What kind of actions do customers take offline? Extract offline channels from #7and use them for customer development.

Online

- Web application
- Mobile devices
- Agricultural forums
- Social media

Offline

- Farmers discussion groups
- Agricultural extension services
- Local farming communities